Dollar Stablecoin and Arbitrage under Capital Controls

Evidence from the Kimchi Premium

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Master's Thesis

Outline

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Kimchi premium

- The kimchi premium refers to price discrepancies between Korean and offshore exchanges.
- Marginal investors outside the US and Europe are willing to pay more for bitcoin in response to positive news or sentiment.

$$Premium^{Kimchi} = \frac{P_{\mathrm{KR}}^{BTC/KRW}}{FX^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1$$

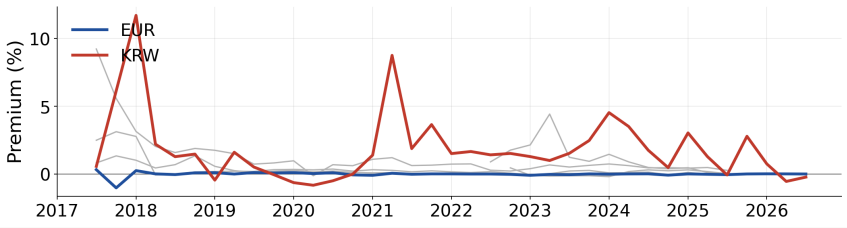
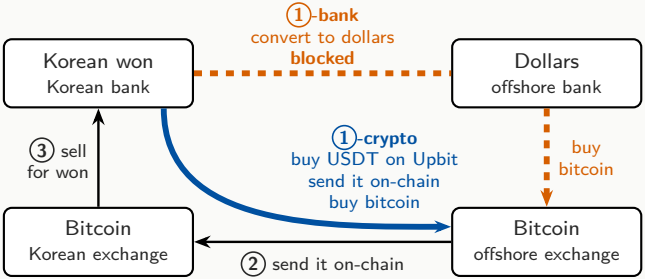


Figure 1: Bitcoin's premium in seven local currencies over Coinbase's USD price

Bitcoin's price on one local exchange per currency over Coinbase's USD price, quarterly mean of daily premia (Makarov & Schoar 2020). Grey: AUD, BRL, GBP, PHP, TRY, to 2025 Q3; KRW and EUR to 2026 Q3.

▸ Capital controls

Two routes for arbitrage: banking system or blockchain network



Makarov & Schoar (2020, *Journal of Financial Economics*)

*"[C]apital controls are most relevant for **the price of crypto to fiat currencies**. (①-**bank**)*
*In contrast, **transactions between two cryptocurrencies** (①-**crypto**) should not be affected [. . .]."*

Paying more for bitcoin, or for dollars?

- **USDT** is a cryptocurrency **pegged 1-to-1 to the US dollar** that **moves through blockchain**.
- Griffin & Shams (2020): stablecoins serve as the **medium for cross-exchange arbitrage**.

$$\underbrace{Premium^{Kimchi}}_{\text{kimchi premium}} \approx \underbrace{Premium^{Dollar}}_{\substack{\text{fiat-to-crypto spread} \\ \textbf{Dollar Demand}}} + \underbrace{Premium^{Crypto}}_{\substack{\text{crypto-to-crypto spread} \\ \textbf{Local Crypto Demand}}}$$

$$\frac{P_{\mathrm{KR}}^{BTC/KRW}}{\boldsymbol{FX}^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1 \approx \frac{P_{\mathrm{KR}}^{USDT/KRW}}{\boldsymbol{FX}^{USD/KRW}} - 1 + \frac{P_{\mathrm{KR}}^{BTC/KRW}}{P_{\mathrm{GLB}}^{BTC/USDT} \cdot \boldsymbol{P}_{\mathbf{KR}}^{\boldsymbol{USDT/KRW}}} - 1$$

Research question and Hypotheses

Research Question

Is USDT an arbitrage vehicle, or excess demand for dollars under capital controls?

H1 USDT not as an arbitrage vehicle on the kimchi premium. ($Premium^{Kimchi}$)
The kimchi premium does not converge if the vehicle is a cryptoasset.

H2 USDT as an arbitraged target on the dollar premium. ($Premium^{Dollar}$)
USDT/KRW traded at a premium to the USD/KRW exchange rate in Korea.

H3 USDT as an arbitrage vehicle on the crypto premium. ($Premium^{Crypto}$)
The crypto premium is negligible; when it emerges, USDT serves as the arbitrage vehicle.

Three strands of literature

- **Limits to arbitrage.** Identical assets can trade at different prices across markets because arbitrageurs face capital constraints and risk.
  Shleifer & Vishny (1997, *Journal of Finance*); Froot & Dabora (1999, *Journal of Financial Economics*); Gromb & Vayanos (2002, *Journal of Financial Economics*)

- **The kimchi premium and capital controls.** Crypto premia are largest across borders with capital controls, which bind the fiat-to-crypto price.
  Makarov & Schoar (2020, *Journal of Financial Economics*); Choi, Lehar & Stauffer (2022, working paper, R&R *Journal of Financial and Quantitative Analysis*)

- **Stablecoins in crypto markets.** Stablecoins serve as a common numeraire and an arbitrage vehicle across crypto exchanges.
  Griffin & Shams (2020, *Journal of Finance*); Kang, Kang, Kim & Sul (2025, *Economics Letters*)

Contribution

1. Decomposing the kimchi premium using USDT/KRW prices reveals that the dollar premium accounts for nearly all of it.
2. USDT's persistent KRW premium is consistent with frictions in international capital mobility despite its advantages: cross-border transferability and price stability.

Empirical Design: Flow Variables

x: cryptoasset, t: time

On-chain

$$NOut_{x,t} = Reserve_{x,t} - Reserve_{x,t+1}$$
$$NIn = -NOut$$

$$\frac{1}{3}\sum_{k=0}^{2} \Delta Prem_{t+k} = \alpha + \boldsymbol{b_j}\, NOut_{j,t+h} + b_{BTC}\, NIn_{BTC,t+h} + \gamma' W_{t-1} + \delta\, 1[FeeDC]_t + \varepsilon_t$$

- Onchain flows denominated in BTC, and filtered over $1,000 by the minute
- Controls: $W_{t-1} = (\sigma_{t-1},\ \Delta Prem_{t-1},\ \Delta Prem_{t-1}\,\sigma_{t-1})'$,
  σ_t: s.d. of $\Delta Prem$ over the 12 hours to t
- $1[FeeDC]_t$: 1 during Upbit's USDT/KRW fee discount event, 0 otherwise.
- Newey-West errors, three lags at maximum

Data: Summary Statistics

Table 1: Summary statistics

	N	Mean	SD	Min	P5	P50	P95	Max	AR(1)
Panel A: Upbit with offshore exchanges									
$NOut_{USDT,t}$	40,210	3.36	17.10	-362.50	-14.95	1.78	24.77	421.77	0.26
$NOut_{XRP,t}$	40,210	0.31	12.83	-616.05	-13.54	0.28	14.64	379.62	0.55
$NOut_{ETH,t}$	40,210	-1.03	9.40	-779.63	-11.45	0.00	6.18	92.50	0.47
$NOut_{DOGE,t}$	40,209	-0.23	9.97	-1347.89	-5.14	0.00	4.94	160.23	0.40
$NIn_{BTC,t}$	40,210	0.36	14.19	-399.85	-12.48	-0.00	17.68	540.00	0.25
Panel B: Upbit with Binance and Bybit									
$NOut_{USDT,t}$	40,200	2.75	15.82	-573.39	-12.00	1.34	21.13	417.42	0.19
$NOut_{XRP,t}$	40,210	0.24	11.11	-557.52	-11.68	0.20	12.81	335.82	0.54
$NOut_{ETH,t}$	40,200	-0.42	6.37	-529.94	-7.85	0.00	5.10	68.42	0.43
$NOut_{DOGE,t}$	40,200	-0.13	8.82	-1281.26	-3.74	0.00	3.87	139.03	0.36
$NIn_{BTC,t}$	40,200	0.01	11.63	-399.81	-10.17	-0.00	11.00	540.00	0.24
Panel C: Upbit with other Korean exchanges									
$NOut_{USDT,t}$	40,210	-2.61	5.86	-121.86	-12.48	-1.04	2.41	154.91	0.48
$NOut_{XRP,t}$	40,210	-0.37	3.02	-112.87	-4.09	-0.02	2.26	63.71	0.41
$NOut_{ETH,t}$	40,210	-0.13	1.42	-66.38	-1.46	0.00	0.82	32.53	0.23
$NOut_{DOGE,t}$	40,209	0.02	0.61	-10.81	-0.04	0.00	0.00	34.90	0.34
$NIn_{BTC,t}$	40,210	0.03	3.12	-68.00	-1.04	-0.00	1.83	171.69	0.19
Panel D: Kimchi premium									
$Premium_t^{Kimchi}$	40,210	101.5	180.0	-482.5	-159.9	87.9	441.3	1073.4	1.00

- Sample: 2024-06-10 – 2026-09-30, from the third day after the USDT/KRW listing, in half-hour bins.
- Excluded: the bins within martial law's hours, 2024-12-03 22:27 – 2024-12-04 04:30 KST, in Tables 1–5.
- Left out of the flows: Tether Treasury transfers and Upbit's internal moves, staking included.

▸ Data sources

Data: Correlation

Table 2: Correlations and autocorrelations of the onchain flows at Upbit

Panel A: $NOut_{x,t}$, pairwise correlations

	BTC	ETH	XRP	SOL	DOGE	USDT TRON	USDT ETH
BTC	1.00						
ETH	0.01	1.00					
XRP	-0.08	0.00	1.00				
SOL	-0.02	0.19	0.08	1.00			
DOGE	-0.07	0.01	0.06	0.17	1.00		
USDT TRON	-0.05	-0.17	-0.25	-0.18	-0.14	1.00	
USDT ETH	-0.03	-0.02	0.03	-0.03	0.00	0.02	1.00

Panel B: $NOut_{x,t}$, autocorrelations

	10 min		30 min		1 hour	
	Coefficient	t-Stat	Coefficient	t-Stat	Coefficient	t-Stat
BTC	+0.100	+10.49	+0.251	+13.65	+0.361	+14.47
ETH	+0.468	+8.36	+0.465	+8.33	+0.477	+6.92
XRP	+0.537	+19.65	+0.547	+19.66	+0.476	+16.33
SOL	+0.342	+16.66	+0.382	+19.18	+0.422	+17.22
DOGE	+0.306	+2.32	+0.405	+2.44	+0.428	+2.70
USDT TRON	+0.269	+18.54	+0.301	+18.87	+0.325	+19.11
USDT ETH	-0.189	-2.56	-0.021	-0.22	-0.143	-1.12

Unlike other cryptoassets, USDT serves as dollar-denominated cash within the exchange.

Results: Kimchi Premium (H1)

Table 3: The effect of flows of BTC and USDT, XRP, ETH and DOGE on the kimchi premium

After USDT/KRW listing	j = USDT		j = XRP	j = ETH	j = DOGE
	(1)	(2)	(3)	(4)	(5)
	10 min	30 min	30 min	30 min	30 min
$NOut_{j,t-1}$	+0.066**	+0.067	+0.176*	+0.291***	+0.056
	(+2.17)	(+0.90)	(+1.71)	(+4.21)	(+0.55)
$NIn_{BTC,t-1}$	-0.105***	-0.292***	-0.299***	-0.281***	-0.293***
	(-4.34)	(-3.90)	(-4.01)	(-3.72)	(-3.95)
σ_{t-1}	-0.015	-0.009	-0.007	-0.004	-0.007
	(-0.95)	(-0.54)	(-0.42)	(-0.23)	(-0.45)
$\Delta Premium_{t-1}$	-0.063***	-0.055***	-0.052***	-0.053***	-0.054***
	(-8.71)	(-8.21)	(-7.63)	(-7.93)	(-8.03)
$\Delta Premium_{t-1}\sigma_{t-1}$	-0.001	-0.001**	-0.001**	-0.001**	-0.001**
	(-1.25)	(-2.02)	(-2.16)	(-2.11)	(-2.06)
$FeeDC_{j,t}$	+0.032	+0.039			
	(+0.69)	(+0.32)			
Constant	+0.119	+0.095	+0.081	+0.043	+0.089
	(+0.92)	(+0.49)	(+0.43)	(+0.22)	(+0.46)
Observations	120,646	40,208	40,208	40,208	40,208
Adjusted R^2	+0.026	+0.023	+0.023	+0.024	+0.023

Withdrawing crypto from Upbit counterintuitively raises $Premium^{Kimchi}$.

Results: Dollar Premium (H2)

Table 4: The effect of flows of USDT on the dollar premium

After USDT/KRW listing

	30 min				
	(1) Issuance	(2) No Issuance	(3) Lag Neg	(4) Lag Pos	(5) Lag Neg, Issuance
$NIn_{USDT,t-1}$	-0.606*	-0.023	-0.219*	-0.056	-0.839**
	(-1.86)	(-0.40)	(-1.95)	(-0.58)	(-2.28)
σ_{t-1}	-0.054	-0.009	-0.005	+0.004	+0.008
	(-1.51)	(-0.40)	(-0.23)	(+0.24)	(+0.25)
$\Delta Premium_{t-1}$	-0.051**	-0.073***	-0.071***	-0.032	-0.133*
	(-2.49)	(-3.46)	(-3.37)	(-1.64)	(-1.86)
$\Delta Premium_{t-1}\sigma_{t-1}$	-0.000	+0.001	+0.001	-0.001**	+0.001
	(-1.14)	(+0.67)	(+1.02)	(-2.37)	(+1.25)
$FeeDC_{j,t}$	+0.545*	+0.014	-0.372***	+0.528***	-0.348
	(+1.67)	(+0.12)	(-2.67)	(+3.79)	(-0.96)
Constant	+0.441	+0.099	+0.184	-0.380	-0.731
	(+1.02)	(+0.41)	(+0.61)	(-1.44)	(-0.89)
Observations	6,028	34,180	18,179	17,780	2,765
Adjusted R^2	+0.023	+0.012	+0.006	+0.015	+0.021

Issuance: within 72 hours after Tether Treasury issuance into Upbit's cold wallet;
Lag Neg $= \Delta Prem_{t-1} < 0$

In contrast, depositing USDT into Upbit exerts downward pressure on $Premium^{Dollar}$ only conditionally:
following new USDT issuance together with a prior decline in the premium
(weakly, at 10%, after either one)

Results: Crypto Premium (H3), suspension of onchain transfers

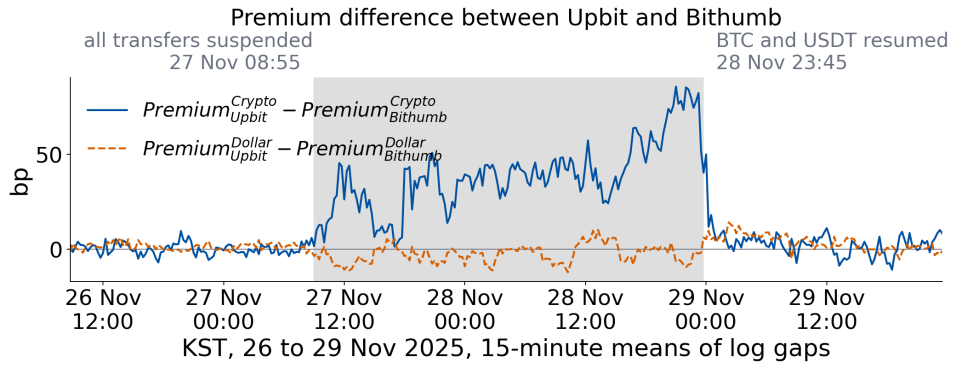


Figure 2: Upbit less Bithumb in the dollar and crypto premia around the suspension

- With suspension, Upbit sellers could sell only what they already held.
- As a result, $Premium^{Crypto}$ gap spikes while $Premium^{Dollar}$ gap stays flat.

▸ Fence pumping

Results: Crypto Premium (H3), Vehicle Flow

Vehicle flows

$$ArbFlow^{Off}_{j,t} = \left(NOut^{Off}_{j,t} + NIn^{Off}_{BTC,t}\right)/2$$

j leaves and BTC arrives at Upbit;

$Off = \{\text{Binance, Bybit}\}$

$$\begin{aligned}\Delta Premium^{Crypto}_{t} = {} & \beta_0 + \beta_1\, \Delta Premium^{Crypto}_{t-1} \\ & + \beta_2\, 1[HighArbFlow]^{Off}_{t-1} + \beta_3\, 1[HighArbFlow]^{Off}_{t-1} \times \Delta Premium^{Crypto}_{t-1} \\ & + \beta_4\, 1[HighVol]_{t-1} + \beta_5\, 1[HighVol]_{t-1} \times \Delta Premium^{Crypto}_{t-1} + \varepsilon_t\end{aligned}$$

Estimated apart for $\Delta Premium^{Crypto}_{t-1} > 0$ and < 0; after a fall, $1[LowArbFlow]$:
the bottom percentiles, BTC leaving and j arriving. $Premium^{Crypto}$ priced through j.

Results: Crypto Premium (H3), Upbit

Table 5: Crypto Premium Reversals and Vehicle Flow

Lagged $\Delta Premium > 0$	j = USDT			j = XRP		
	Top 1%	Top 5%	Top 10%	Top 1%	Top 5%	Top 10%
$\Delta Premium_{t-1}$	-0.4311***	-0.4307***	-0.4194***	-0.4274***	-0.4062***	-0.4134***
	(-36.67)	(-28.62)	(-24.75)	(-27.59)	(-28.96)	(-27.25)
$1[HighArbFlow]^{Off}_{t-1}$	+4.2806**	+1.7184	+1.5787*	-1.8150*	+0.8764	+0.2105
	(+2.45)	(+1.17)	(+1.65)	(-1.70)	(+0.93)	(+0.30)
$1[HighArbFlow]^{Off}_{t-1} \times \Delta Premium_{t-1}$	-0.7417***	-0.4031*	-0.3625**	-0.2108*	-0.4218***	-0.3239***
	(-3.41)	(-1.89)	(-2.53)	(-1.69)	(-3.39)	(-3.40)
$1[HighVol]_{t-1}$	-1.8477	-1.2286	-0.9265	-0.7113	-1.2125	-0.7344
	(-0.84)	(-0.86)	(-0.83)	(-0.35)	(-1.40)	(-1.06)
$1[HighVol]_{t-1} \times \Delta Premium_{t-1}$	+0.2481	+0.1986	+0.1840	+0.2118	+0.1718**	+0.1700***
	(+1.62)	(+1.41)	(+1.52)	(+1.34)	(+2.33)	(+2.78)
$FeeDC_{j,t}$	-0.0841	-0.0622	-0.0481			
	(-0.84)	(-0.62)	(-0.47)			
Constant	-0.1662*	-0.1645	-0.2499**	-0.1661	-0.2590**	-0.1690
	(-1.80)	(-1.47)	(-2.01)	(-1.40)	(-2.43)	(-1.45)
Adjusted R^2	0.091	0.087	0.088	0.085	0.092	0.094
Observations	19,971	19,971	19,971	19,703	19,703	19,703

USDT's offshore round trip reverses the premium's prior move in both directions; XRP's only after a rise.

▸ Lagged $\Delta Premium < 0$

Conclusion, discussion and policy implications

Policy implications

1. Listing dollar stablecoin at local fiat market does not ensure that its price tracks exchange rate.
2. The kimchi premium is primarily the shadow cost of obtaining dollars through USDT, rather than a crypto-specific price gap.
3. Capital controls force arbitrageurs to hold bitcoin in Korea in advance, exposing them to price risk.

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Appendix

Data Sources

12 crypto assets (BTC, ETH, XRP, USDT, . . .) on **Upbit** and **Bithumb**, against **Binance**.

Table A1: Data sources

Data	Frequency	Provider	Coverage	Period
Prices	Hourly, minute	Upbit, Bithumb, Binance, Bybit	12 assets, KRW and USDT	2017-10-24 – 2026-09-25
FX rate	Minute-level	Barchart	USD/KRW interbank	2017-10-24 – 2026-09-25
Trades	Tick-level	Upbit	35 books (BTC, USDT, . . .)	2022-05-01 – 2026-09-25
Holdings	Daily	Arkham Intelligence, XRP Ledger, Hyperion, Polkadot, Cardano, Polygon	15 assets, both venues (DOT, ADA: Upbit)	2017-10-24 – 2026-09-25
Transfers	Transaction-level	Arkham Intelligence, XRP Ledger, Blockchain.com, TronGrid, Tenderly	6 assets, 15.6M transfers	2019-02-28 – 2026-09-25
Notices		Upbit, Bithumb	669 assets, 2,813 notices	2017-12-12 – 2026-09-25

◂ Summary Statistics

Table A2: Variable definitions

Variable	Definition
$Premium_t^{Kimchi}$	$P_t^{BTC/KRW} \,/\, \left(FX_t^{USD/KRW} \cdot P_{GLB,t}^{BTC/USDT}\right) - 1$
$Premium_t^{Dollar}$	$P_t^{USDT/KRW} \,/\, FX_t^{USD/KRW} - 1$
$Premium_{j,t}^{Crypto}$	$P_t^{BTC/KRW} \,/\, \left(P_{GLB,t}^{BTC/USDT} \cdot P_t^{j/KRW} / P_{GLB,t}^{j/USDT}\right) - 1$
$\Delta Reserve_{x,t}$	$Reserve_{x,t} - Reserve_{x,t+1}$
$NOut_{x,t}$, $NIn_{x,t}$	$Out_{x,t} - In_{x,t} = \Delta Reserve_{x,t}, \quad NIn_{x,t} = -NOut_{x,t}$
$ArbFlow_t$	$\left(\sum_{m=t-1}^{t} NOut_{USDT,m} + \sum_{m=t-1}^{t} NIn_{BTC,m}\right)/2,$
$NOut_{USDT,m}$, $NIn_{BTC,m}$	$Out_{USDT,m} - In_{USDT,m}, \quad In_{BTC,m} - Out_{BTC,m}$
Vol_{t-1}	standard deviation of $\Delta Premium^{Crypto}$ over the 12 hours before t

Crypto Share of Upbit's Trading Value

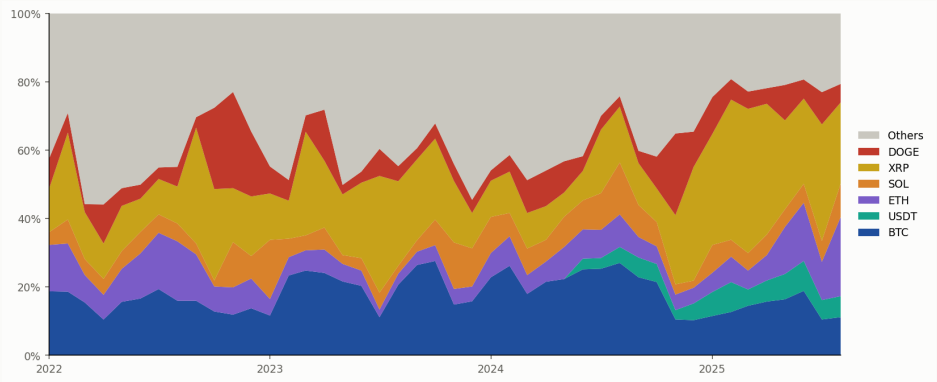


Figure A1: Crypto share of Upbit's monthly KRW trading value, Jan. 2022 – Aug. 2025

Months in UTC, from the one-minute candles, price × volume. Others: the other KRW books held; the figure stops in Aug. 2025, the last month before Upbit's books held fall from 36 to 18.

Korea won's capital controls

Table A3: Korea's capital controls

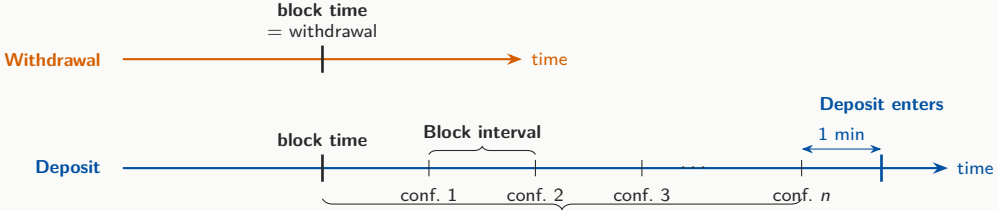
Rule	In force	Source
Remittance cap	A resident may send abroad without documents at most **USD 50,000** a year (1999 to 3 July 2023), **USD 100,000** a year since 4 July 2023, and **USD 5,000** per transfer (from Jan 2026)	Ministry of Economy and Finance (10 Feb 2023; 8 Dec 2025)
Exchange access	Won deposits require a real-name bank account; **non-residents cannot trade** on Korean exchanges (since 30 Jan 2018)	Financial Services Commission (23 Jan 2018)
Enforcement	**USD 12.26 billion** of outward remittances by 84 firms, collected from crypto exchanges and sent abroad as import payments, 2021–22	Financial Supervisory Service (4 Apr 2023)

◂ Kimchi premium

When Each Transfer Enters the Flows

Table A4: Upbit's crypto transfer confirmation timing

	Upbit confirmations	Block interval	Deposit enters
BTC	1	10 min	next block's time + 1 min
ETH	36	12 s	36th block on top's time + 1 min
USDT (Tron)	1	3 s	block time + 1.05 min
USDT (Ethereum)	36	12 s	36th block on top's time + 1 min
XRP	1	4 s	block time + 1.07 min
DOGE	6	1 min	6th block on top's time + 1 min
SOL	1	0.4 s	block time + 1.01 min



Upbit confirmations n: blocks mined on top (BTC, DOGE, ETH: the confirming block's own time)

Upbit's `minimum_deposit_confirmations`, read 21 Oct. 2026; every NIn and NOut in Tables 1–4 at the time shown; Tron, XRP, SOL and ETH before June 2024 (no block numbers): confirmations × block interval. Left out: Upbit's internal moves (staking), Tether Treasury transfers, 3 Dec. 2024 (martial law).

How the Flows Are Built

- **USDT** sums both of Upbit's networks, Tron and Ethereum, in one series. Tether Treasury's transfers on Tron are left out.
- **Timing:** each transfer enters at its own time. A withdrawal enters at its block, a deposit one minute after its network's Upbit confirmations (Tron USDT: 1 block of 3 s; Ethereum USDT, as ETH: 36 blocks, the 36th at its own time).
- **BTC terms:** a transfer is valued at its crypto's KRW price over the BTC/KRW price of its entry minute (BTC's own at its amount). Transfers are summed per minute, a minute under $1,000 counts as zero, and the minutes are summed into the half-hour bin; a bin holding a transfer with no price yet is missing.
- **14-month blocks:** Tables 3 and 4 standardize each flow within blocks of 14 months counted from the sample's first bin, the remainder a block of its own. Table 5 sets its top 1, 5 and 10% cutoffs within the same blocks.
- **Martial law:** the bins within its hours, 2024-12-03 22:27 – 2024-12-04 04:30 KST (13:27 – 19:30 UTC), are left out of Tables 1–5, and so are the bins whose lag, lead or 12-hour Vol_{t-1} would reach into them. The rest of that day stays in.

▸ Confirmation timing

Tether Treasury Pays Upbit Directly

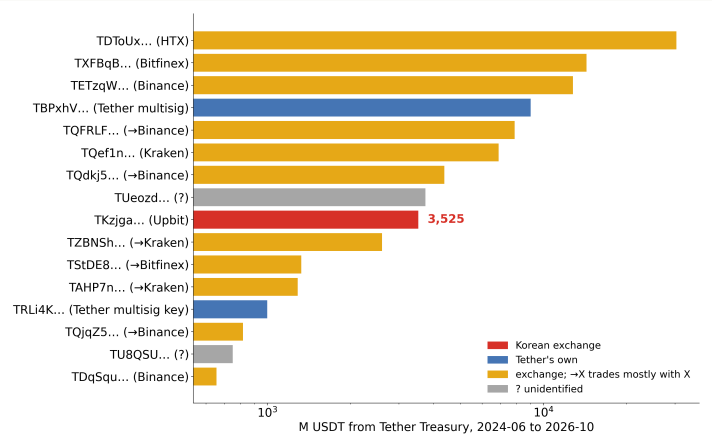


Figure A2: Every Tron address Tether Treasury paid 500M USDT or more from June 2024, log scale

Source of Labells: Tronscan tags, Whale Alert and Upbit's own wallets.

→X: unnamed, at least half its other USDT is with exchange X's labelled wallets (its latest 1,000 transfers).

Fence pumping after Upbit's 2025 hack



TV CHOSUN 뉴스 ● 라이브

경제 > 전체

업비트, 입출금 순차적 재개…'가두리 펌핑'에 수수료 수익은 되레 증가

이정연 기자 | 경제부 등록: 2025.12.01 오후 20:47 • 수정: 2025.12.01 오후 21:21

국내 최대 가상자산 거래소 업비트가 해킹 사고에도 하루 만에 수익 원의 수수료를 벌어들였다.

업비트가 지난 27일 445억 원 규모의 해킹 사고 직후 일부 가상화폐 입출금을 중단하면서 이른바 '가두리 펌핑'이 발생해 거래량이 늘어났기 때문이다.

거래소 간 코인 이동이 막히자 업비트 안에서 특정 종목에 매수세가 쏠려 시세가 급등한 것이다.



연합뉴스

두나무-네파 합병 회견날, 업비트서 445억 규모 해킹 사고(종합2보)

송고 2025-11-27 16:10

오 대표는 "회원 자산에 어떤 피해도 발생하지 않도록 전액 업비트 자산으로 충당할 예정"이라고 말했다.

- Upbit **temporarily suspended deposit and withdrawal services** after hacking.
- With suspension, buying pressure concentrated on several cryptos and **pushed their prices up**: 'fence pumping'.

Sources: Yonhap News, 2025-11-27 16:10; TV Chosun, 2025-12-01 20:47. 'Fence pumping': *gaduri pumping*.

Crypto premium

Appendix: Crypto Premium (H3), Lagged $\Delta Premium < 0$

Table A5: Crypto Premium Reversals and Vehicle Flow

Lagged $\Delta Premium < 0$	j = USDT			j = XRP		
	Bottom 1%	Bottom 5%	Bottom 10%	Bottom 1%	Bottom 5%	Bottom 10%
$\Delta Premium_{t-1}$	-0.4723***	-0.4487***	-0.4310***	-0.4336***	-0.4555***	-0.4547***
	(-38.99)	(-35.76)	(-28.20)	(-19.76)	(-27.32)	(-26.26)
$1[LowArbFlow]^{Off}_{t-1}$	-4.2762***	-2.8389***	-2.2647***	+1.4383	+2.5734*	+1.3887
	(-3.36)	(-4.14)	(-3.57)	(+0.58)	(+1.76)	(+1.61)
$1[LowArbFlow]^{Off}_{t-1} \times \Delta Premium_{t-1}$	-0.7138***	-0.5563***	-0.4524***	-0.1674	-0.0008	-0.0655
	(-5.09)	(-5.80)	(-4.92)	(-0.71)	(-0.00)	(-0.58)
$1[HighVol]_{t-1}$	+0.7659	+0.5109	-0.1257	-0.5679	-0.0820	+0.2012
	(+0.45)	(+0.63)	(-0.19)	(-0.25)	(-0.08)	(+0.27)
$1[HighVol]_{t-1} \times \Delta Premium_{t-1}$	+0.2334	+0.1323	+0.0457	+0.0681	+0.0636	+0.0861
	(+1.58)	(+1.54)	(+0.61)	(+0.45)	(+0.68)	(+1.19)
$FeeDC_{j,t}$	+0.0302	+0.0239	+0.0093			
	(+0.35)	(+0.28)	(+0.11)			
Constant	-0.0852	+0.0532	+0.1635	+0.1044	-0.1196	-0.1474
	(-0.90)	(+0.55)	(+1.45)	(+0.67)	(-0.98)	(-1.15)
Adjusted R^2	0.170	0.168	0.165	0.093	0.094	0.095
Observations	20,130	20,130	20,130	19,911	19,911	19,911

◂ Crypto premium

Bitcoin's Price and Its Hodrick–Prescott Trend

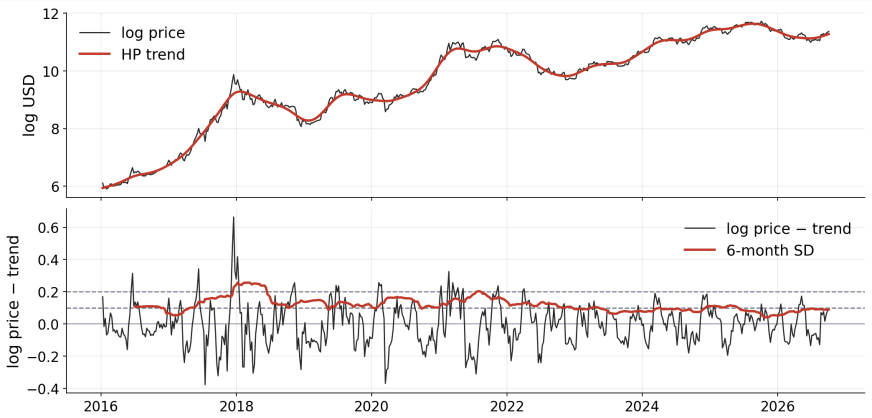


Figure A3: Hodrick–Prescott trend (top) and deviation from it with its 6-month (26-week) SD (bottom), Jan. 2016 – Oct. 2026

The lower peak of $Premium^{Kimchi}$ might result from the lower volatility of the offshore bitcoin price.

$\lambda = 1600$. Makarov & Schoar (2020), Fig. 7, take the deviation as buying pressure over Jan. 2017 – Apr. 2018.